

Enhancement Three Narrative: Databases

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Artifact Description

The artifact I selected for the databases category is the same text-based adventure game that I enhanced in the previous milestones. The original version of this project was created early in my Computer Science program and did not include any form of data storage. For this milestone, I added a database feature to the game so it can record the results of each playthrough. This allowed me to integrate actual database skills into the project and show that I can work with structured data in a real program.

Justification for Inclusion

I included this artifact in my ePortfolio for the database category because the new enhancement adds a clear and useful database component. I used SQLite to store important information at the end of every game session, such as whether the player won or lost, how many gemstones they collected, how many moves they made, and the timestamp of the game. This information is saved into a database table called `castle_game_results`.

This enhancement shows my ability to connect a Python program to a database, create tables, insert data, and manage a permanent record of game sessions. Even though the game is small, it now has a way to store and track past results, which makes it feel more complete. It also shows that I can use SQL commands and handle stored data in a structured and organized way. I also made sure the database writes were safe and that the program properly closes the connection when the game ends.

Course Outcome Connection

In Module One, I planned to meet the course outcome related to database skills with this enhancement. I believe I met this goal because I used a real database system, wrote SQL

statements, and designed a table that fits the needs of the program. I also used Python's database library to connect the game to the database and store information in it. The enhancement shows that I can solve a data storage problem using database tools and that I can integrate database logic into an existing software project. I do not need to update my outcome plan because this enhancement fits exactly what I intended.

Reflection on the Enhancement Process

Adding a database to the game helped me understand how data storage can support the rest of the program. Before this enhancement, the game ended and all information about the player's progress disappeared. After adding the database, the game can now record important details from each playthrough. This reminded me about just how many software applications depend on databases to store user actions, results, or history.

One challenge I faced was deciding what information should be stored. I needed to choose data that made sense for this type of game and would be useful if someone wanted to analyze or review past playthroughs. Another challenge was making sure the database connection opened and closed correctly, especially in cases where the game might end early or unexpectedly. Working through these problems helped me improve my understanding of how applications manage persistent data.

Overall, this enhancement taught me how to design a simple database table, write SQL statements, and store real program output in a reliable way. It also helped me connect the idea of data storage to a practical example, which is an important skill for software development. This is why this artifact, with the new database feature, belongs in my ePortfolio.